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ADRC-based Longitudinal Control for Novel Flight Vehicle with Moving Mass

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***Abstract*—An adaptive longitudinal control law for moving mass control of a novel flight vehicle configuration is proposed. The coupling dynamic model between angle of attack and deflection of moving mass is generated. The dynamic analysis indicates that the proposed moving mass configuration with larger mass ratio gives rise to more dynamic coupling and maneuverability. Moreover, the control authority is determined by the mass ratio of the moving mass and the difference between the mass center of the moving mass and the mass center of the vehicle body. To deal with the coupling caused by the additional inertia moment of moving mass, the coupling dynamics system is transformed into a cascade system for controller design. Active disturbance rejection control framework is employed to design the adaptive longitudinal controller. Its extended state observer estimates the total disturbance to overcome the uncertainties in the flight vehicle model. The simulation results show that the proposed configuration is feasible and validate the quality of the adaptive controller which ensures good performance.**

I. Introduction

Moving mass control technology has been applied to many fields as a control methodology [1-3], where the internal moving mass changes the relative location of its body with respect to the vehicle body to effect a change of attitude. Recently, this control technology was used for the attitude control of reentry vehicles. It has several advantages compared with conventional aerodynamic control surfaces and reaction control systems (RCS). The research of the reentry vehicle with moving mass focuses on the configuration of the internal moving mass, the dynamic modeling, the dynamic analysis, and the control system design.

Different configurations of internal moving masses generate different control mechanisms. A configuration with two or three moving masses has been applied in kinetic warheads [4][5]. In these configurations, the moving mass offers the capacity of longitudinal and lateral maneuvering. Although the configuration with multiple moving masses shows excellent and various maneuvers, it is difficult to implement for engineering because more actuators are contained within a limited space of the vehicle body for driving the moving masses. The single moving-mass control system, which is the simplest configuration, provides roll control ability [6], or generates a trim angle of attack (AOA) for maneuvering [7]. This control strategy is easy to implement and applicable, but its control authority is limited.

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In order to improve the maneuver of the single moving mass flight vehicle, some novel configurations have been developed. Gao [8] develops a system that the moving mass rail can rotate with respect to the roll axis of the body so that the vehicle can track pitch and yaw channel commands. Wei [9] proposed a combination of bank-to-turn control with the single moving mass and reaction jets to control the pitch and roll channel. To implement the moving mass control technology in engineering further, this paper presents a novel configuration with a large mass ratio moving mass. This configuration could improve the maneuver performance and optimize the internal space of the flight vehicle.

Previous research has demonstrated that the controller of using the linear model is successfully applied to moving mass system. Rogers [10] investigated the control authority and stability for the linear model about the AOA and the sideslip angle, which is obtained by neglecting the nonlinear terms and perturbation of the dynamic model. Frost [11] established a modified projectile linear theory and investigated the frequency and damping properties for the epicyclic dynamics. Wang [12] derived the transfer function of roll channel by simplifying the nonlinear model and designed a controller via the standard coefficient method for a single moving-mass asymmetrical reentry vehicle. Although the simplified linear model is convenient to design the controller of the moving mass system, difficulties arise as a result of various uncertainties in the aerodynamic parameters and nonlinear nature of the system dynamics with the increase of mass ratio of the moving mass. It is necessary to develop a controller to overcome uncertainties in the nonlinear dynamics, nonlinear adaptive control methods have been developed such as classical adaptive control [13], adaptive backstepping method with neural network [14], and L_1 adaptive control [15]. These approaches give a certainty-equivalent adaptive law, but its estimation shows a poor performance, which may cause the undesired response of the closed-loop system [16].

The active disturbance rejection control (ADRC) technology, which was proposed by Han [17], originally adopted the nonlinear extended state observer (ESO) and the nonlinear control law. This control method can reject disturbance and require very little information of the plant dynamics, but its large number of tuning parameters and complex structure increase the design difficulty. Gao [18] proposed a tuning method based on bandwidth concept to simplify the design process. He also developed the linear extended state observer (LESO) to improve the observer structure [19]. ADRC offers a solution where the necessary modeling information needed for the feedback control system to function well is obtained through the input–output data of the plant in real time. It has been applied in many practical systems, such as fuel cell power system [20], the jet engine

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problem [21], the selected catalytic reduction system [22]. These applications of ADRC demonstrate its enormous potential for control engineering.

The influence of dynamic behavior of the moving mass becomes serious with the increase of the mass ratio. The dynamical coupling between the body attitude and the actuator mass and the servo loop control of actuator mass must be considered. In this paper, for the proposed configuration of the moving mass with large mass ratio, the coupled dynamics model of the body attitude and the actuator mass are investigated first. A coupling index is investigated to reveal the coupling characteristics and evaluate the coupling effects of the moving mass. Moreover, the effect of moving mass parameters on the control authority is examined. In general, the moment of inertia of the moving mass was supposed to be neglected or a perturbation for decoupling the attitude-servo control system [5][8]. But we transform the coupling system into a cascade system and presented an adaptive controller based on ADRC for the longitudinal dynamics of the moving mass flight vehicle. To overcome the uncertainties in the flight vehicle model, the unknown dynamics and external disturbances are regarded to be an augmented state of the system and are estimated using the ESO. Simulation results show that the proposed adaptive controller can ensure good performance in the novel configuration with internal moving mass.

II. Dynamics Modeling

A. Sketch of moving mass flight vehicle

The point moving mass control configuration [9] has two difficulties in engineering: 1) the point moving mass cannot satisfy the desired mass in small size, 2) the moving mass and actuator occupy the main internal space, which increases the design difficulty of internal configuration.

To solve the above problems, a sketch of the moving mass flight vehicle is proposed in Fig. 1, which is characterized by using single moving mass with large mass ratio in combination with jet thrusters to achieve bank-to-turn control mode. It consists of three major components, namely, a main vehicle shell, an internal single moving mass that can move along an internal rail, and a reaction control system (RCS). The moving mass is fixed with the vehicle shell at O, while the other side C can translate along the rail under the action of servo force, and the deflection angle of moving mass with respect to the longitudinal axis of vehicle shell is denoted as δ. The mass center of the moving mass and vehicle shell are denoted by p and b, and s denotes the mass center of the entire system. The mass center offset caused by the moving mass can generate a trim AOA resulting from aerodynamic drag. The roll channel is controlled by the jet thrusters to achieve the desired orientation.

m_P and m_B are the masses of the moving mass P and the body B, respectively. The mass of entire system is $m_W = m_P + m_B$. $\mu_P = m_P / m_W$, $\mu_B = m_B / m_W$ are the mass ratios of the moving mass and body mass relative to the system, respectively. L_P and L_B are the distance of the center of mass of the moving mass and center of mass of the body from O. L_{oc} is the distance of the rail from O.

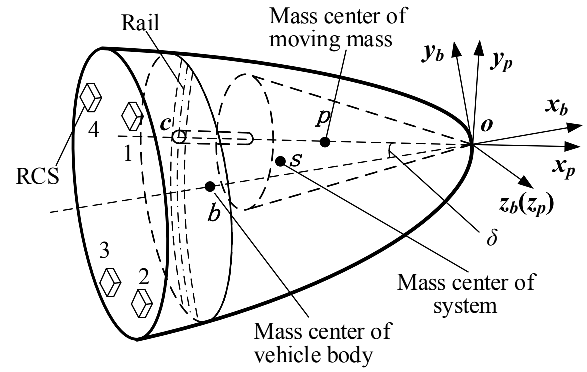

Fig. 1 Proposed moving mass configuration

In the proposed configuration, the internal moving mass generates the angle of attack α by the motion in the longitudinal symmetry plane. Thus, the AOA tracking problem is our concern and the control model must be investigated. The dynamic model about the AOA and the deflection angle are given by [23]

$$\begin{cases} \tilde{I}_{z1}\ddot{\alpha} + \tilde{I}_{\delta 1}\ddot{\delta} = \left(C_z - \tilde{I}_{z1}C_\alpha\right)\dot{\alpha} + \mu_P L_P C_x qS\delta \\ \qquad + \left(m_z^\alpha qSL + (L_P - L_B)\mu_P C_y^\alpha qS + C_z C_\alpha\right)\alpha \\ \tilde{I}_{z2}\ddot{\alpha} + \tilde{I}_{\delta 2}\ddot{\delta} = FL_{oc} - \tilde{I}_{z2}C_\alpha\dot{\alpha} + \mu_P L_P C_y^\alpha \alpha qS + \mu_P L_P C_x qS\delta \end{cases} \quad (1)$$

where

$$\begin{aligned} &\tilde{I}_{z1} = I_{B3} + I_{P3} + m_P\mu_B(L_P^2 + L_B^2 - 2L_P L_B) \\ &\tilde{I}_{\delta 1} = I_{P3} + m_P\mu_B(L_P^2 - L_P L_B) \\ &\tilde{I}_{z2} = I_{P3} + m_P\mu_B(L_P^2 - L_P L_B),\ \tilde{I}_{\delta 2} = I_{P3} + m_P\mu_B L_P^2 \\ &C_\alpha = C_y^\alpha qS / m_s V,\ C_z = m_z^{\omega_z} qSL^2 / V \end{aligned}$$

F is servo force to drive moving mass and $M = FL_{oc}$. q is the dynamic pressure. S is the cross-sectional area, L is the reference length, and V is the magnitude of flight vehicle velocity. C_x is drag aerodynamic coefficient. C_y^α is the partial derivatives of the aerodynamic coefficient with respect to the angle of attack. m_z^α is the partial derivatives of the pitching moment coefficient with respect to the AOA. $m_x^{\omega_x}$ is the aerodynamic damping coefficient.

B. Dynamic coupling and control authority

Equation (1) describes the attitude-servo system of a moving mass flight vehicle, which takes the AOA and servo force as output and control input respectively. According to the above equations, it can be observed that the pitch channel and the deflection angle are coupled by inertia.

For the dynamic coupling, we focus on the angular acceleration relationship between the pith angle and the deflection angle. Thus, factoring $\dot{\omega}_z$ which described in [23] and we obtain

$$\begin{aligned} \dot{\omega}_z &= \tilde{I}_{z1}^{-1}\left[\left(m_z^\alpha \alpha + \frac{m_z^{\omega_z}\omega_z L}{V}\right)qSL \right. \\ &\qquad \left. + (L_P - L_B)\mu_P C_y^\alpha \alpha qS + \mu_P L_P C_x qS\delta\right] - \tilde{I}_{z1}^{-1}\tilde{I}_{\delta 1}\ddot{\delta} \quad (2) \\ &= w_m - \tilde{I}_{z1}^{-1}\tilde{I}_{\delta 1}\ddot{\delta} \end{aligned}$$

Let $\dot{\bar{\omega}}_z = \dot{\omega}_z - w_m$, Eq.(2) rewrite as

$$\dot{\bar{\omega}}_z = -\tilde{I}_{z1}^{-1}\tilde{I}_{\delta 1}\ddot{\delta} \tag{3}$$

The acceleration $\dot{\bar{\omega}}_z$ can be viewed as a virtual acceleration of the pitch channel, generated by the acceleration of the moving mass and aerodynamic moment. To quantify the coupling between the deflection angle and the pitch channel, we define the acceleration coupling index as

$$\rho_z(\delta) = \left| -\frac{I_{\delta 1}}{I_{z1}} \right| \tag{4}$$

The inertia of the moving mass and vehicle determine the index, which indicates the extent of coupling between the deflection angle and the pitch channel.

The control authority of the moving mass determines the maneuverability of the flight vehicle. To analysis the control authority, the steady-state trim angle of attack α_{trim} can be obtained from the first line of Eq.(1):

$$\alpha_{trim} = \frac{\mu_P L_P C_x}{\mu_P (L_B - L_P) C_y^{\alpha} - m_z^{\alpha} L}\delta \tag{5}$$

Note that the deflection angle of the moving mass could generate the trim AOA. The attitude changes with a small static margin relies on a center of mass offset to create a trim AOA resulting from aerodynamic drag. The mass center difference between the mass center of moving mass and the mass center of vehicle body is defined as $\Delta_{BP} = L_B - L_P$. Obviously, μ_P and Δ_{BP} are important designed parameters to determine the control authority.

III. Longitudinal Controller Design

For controller design, the foremost work is decoupling the AOA α and deflection angle δ. Hence, Eq.(1) is transformed to

$$\begin{cases} \ddot{\alpha} = (J_2 C_1 - J_1 C_2)\alpha + (J_2 - J_1) C_3 \delta \\ \qquad\qquad + (J_2 C_z - C_\alpha)\dot{\alpha} - J_1 M \\ \ddot{\delta} = (J_3 C_2 - J_1 C_1)\alpha + (J_3 - J_1) C_3 \delta - J_2 C_z \dot{\alpha} + J_3 M \end{cases} \tag{6}$$

where

$$C_1 = m_z^{\alpha} qSL + (L_P - L_B)\mu_P C_y^{\alpha} qS + C_z C_\alpha$$

$$C_2 = \mu_P L_P C_y^{\alpha} qS,\ C_3 = \mu_P L_P C_x qS$$

$$J_1 = \frac{\tilde{I}_{\delta 1}}{\tilde{I}_{\delta 2}\tilde{I}_{z1} - \tilde{I}_{\delta 1}\tilde{I}_{z2}} = \frac{\tilde{I}_{z2}}{\tilde{I}_{\delta 2}\tilde{I}_{z1} - \tilde{I}_{\delta 1}\tilde{I}_{z2}}$$

$$J_2 = \frac{\tilde{I}_{\delta 2}}{\tilde{I}_{\delta 2}\tilde{I}_{z1} - \tilde{I}_{\delta 1}\tilde{I}_{z2}},\quad J_3 = \frac{\tilde{I}_{z1}}{\tilde{I}_{\delta 2}\tilde{I}_{z1} - \tilde{I}_{\delta 1}\tilde{I}_{z2}}$$

It can be seen that the control input M has a direct effect on the dynamics of both variables α and δ. To reduce the influence of the control input, the change of variable is introduced

$$x_2 = \dot{\alpha} + \frac{J_1}{J_3}\dot{\delta} = \dot{\alpha} + J_{13}\dot{\delta} \tag{7}$$

Then, the dynamic equation of the new state variable can be written as

$$\begin{aligned} \dot{x}_2 = {} & \frac{(J_2 J_3 - J_1^2)C_1}{J_3}\alpha + \frac{(J_2 J_3 - J_1^2)C_3}{J_3}\delta \\ & + \frac{(J_3^2 - J_1 J_2)C_z - J_3 C_\alpha}{J_3}\dot{\alpha} \end{aligned} \tag{8}$$

Hence, we define the new state vector having the origin as a desired equilibrium point as

$$\boldsymbol{x} = [\alpha,\ \dot{\alpha} + J_{13}\dot{\delta},\ \delta,\ \dot{\delta}]^T \tag{9}$$

The dynamics become

$$\begin{cases} \dot{x}_1 = x_2 - J_{13}x_4 \\ \dot{x}_2 = \frac{(J_2 J_3 - J_1^2)C_1}{J_3}x_1 + \frac{(J_2 J_3 - J_1^2)C_3}{J_3}x_3 \\ \qquad + \frac{(J_3^2 - J_1 J_2)C_z - J_3 C_\alpha}{J_3}(x_2 - J_{13}x_4) \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = (J_3 C_2 - J_1 C_1)x_1 + (J_3 - J_1)C_3 x_3 \\ \qquad + J_3(C_2 - C_1)\alpha_c + J_2 C_z (x_2 - J_{13}x_4) + J_3 M \end{cases} \tag{10}$$

It is noted that the control input M acts directly on the state variable x_4, and acts implicitly on other state variables via x_4. The controller aims to make the AOA follow a reference command, then the target behavior of the moving mass system corresponds to $\tilde{\alpha} = \dot{\tilde{\alpha}} = 0$, which is equivalent to $x_1 = x_2 = 0$.

Note that the system (10) naturally decomposes into a cascade connection of three independent subsystem which can be shown as

$$\dot{x}_1 = f_1(x_1, w_1) + x_2 \tag{11}$$

$$\dot{x}_2 = f_2(x_1, x_2, w_2) + b_2 x_3 \tag{12}$$

$$\begin{cases} \dot{x}_3 = x_4 \\ \dot{x}_4 = f_3(x_1, x_2, x_3, w_3) + b_3 u \end{cases} \tag{13}$$

where w_i is uncertain factor. In the first subsystem, the state variable x_2 is regarded as the virtual control input u_1 used to control the output track the desired signal. Then, the virtual control u_1 is the track object of the second subsystem with the corresponding virtual control u_2 (state variable x_3). Finally, the second order subsystem generates the actual control input u to track the virtual control u_2. The control system with three ADRCs is shown in Fig.2.

The method of ADRC includes the nonlinear ESO and the nonlinear state error feedback (NLSEF) control law in each subsystem. For the first-order subsystem, the second-order ESO is proposed to estimate the total disturbance in the following

$$\text{ESO1}: \begin{cases} e_{10} = z_{11} - x_1 \\ \dot{z}_{11} = z_{12} - \beta_{11}e_{10} + u_1 \\ \dot{z}_{12} = -\beta_{12}\,\mathrm{fal}(e_{10}, \lambda_{10}, \sigma_{10}) \end{cases} \tag{14}$$

where β_{11} and β_{12} are positive observer gains and fal(·) is nonlinear function (see [17]). When the gains are appropriately chosen, the variables z_{11} and z_{12} of the ESO are approximated as x_1 and f_1 in real time, respectively.

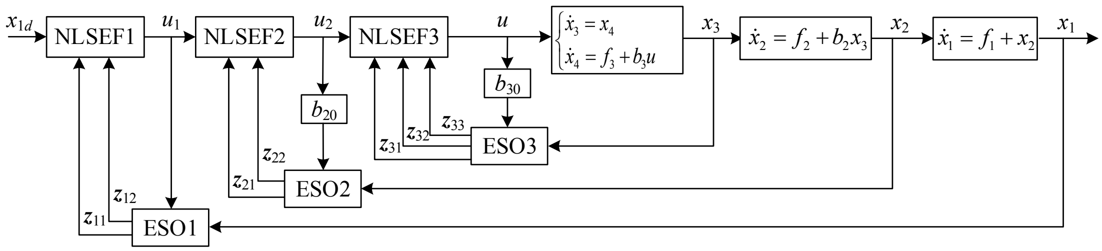

Fig. 2 Control structure of ADRC

The virtual control law can be expressed as follow

$$\text{NLSEF1}:\begin{cases} e_1 = x_{1d} - z_{11} \\ u_1 = x_{2d} = k_1 \text{fal}(e_1, \lambda_1, \sigma_1) - z_{12} \end{cases} \tag{15}$$

where k_1 is positive control gain.

In view of the actual flight environment, there exists perturbation in the aerodynamic parameters. Thus, in the second subsystem, the dynamic term f_2 and gain b_2 are not precise with aerodynamic uncertainty. Let b_{20} be nominal value. Then, the total disturbance of the second subsystem can be defined as

$$d_1 = f_2(x, w_2) + (b_2 - b_{20}) x_3 \tag{16}$$

Eq.(12) can be rewrite as

$$\dot{x}_2 = d_1 + b_{20} x_3 \tag{17}$$

The second-order ESO is designed for

$$\text{ESO2}:\begin{cases} e_{20} = z_{21} - x_2 \\ \dot{z}_{21} = z_{22} - \beta_{21} e_{20} + b_{20} u_2 \\ \dot{z}_{22} = -\beta_{22} \text{fal}(e_{20}, \lambda_{20}, \sigma_{20}) \end{cases} \tag{18}$$

where β_{21} and β_{22} are positive observer gains. Similarly, the variables z_{21} and z_{22} of the ESO are approximated as x_2 and d_1 in real time, respectively. The virtual control law can be expressed as follow

$$\text{NLSEF2}:\begin{cases} e_2 = x_{2d} - z_{21} \\ u_2 = x_{3d} = b_{20}^{-1}[k_2 \text{fal}(e_2, \lambda_2, \sigma_2) - z_{22}] \end{cases} \tag{19}$$

where k_2 is positive control gain.

For the last subsystem, there also exists aerodynamic uncertainty in the dynamic term f_4 and control gain b_4. Let b_{40} be nominal value. The total disturbance of the third subsystem can be defined as

$$d_2 = f_3(x, w_3) + (b_3 - b_{30}) u \tag{20}$$

Eq.(13) can be rewrite as

$$\begin{cases} \dot{x}_3 = x_4 \\ \dot{x}_4 = d_2 + b_{30} u \end{cases} \tag{21}$$

Then a third-order ESO for systems is proposed in the following

$$\text{ESO3}:\begin{cases} e_{30} = z_{31} - x_3 \\ \dot{z}_{31} = z_{32} - \beta_{31} e_{30} \\ \dot{z}_{32} = z_{33} - \beta_{32} \text{fal}(e_{30}, \lambda_{30}, \sigma_{30}) + b_{30} u \\ \dot{z}_{33} = -\beta_{33} \text{fal}(e_{30}, \lambda_{30}, \sigma_{30}) \end{cases} \tag{22}$$

where β_{31}, β_{32}, and β_{33} are positive observer gains. The variables z_{31}, z_{32}, and z_{33} of the ESO are approximated as x_3, x_4, and d_2 in real time, respectively. The actual control law

$$\text{NLSEF3}:\begin{cases} e_3 = x_{3d} - z_{31} \\ e_4 = \dot{x}_{3d} - z_{32} \\ u = b_{30}^{-1}[\text{fhan}(e_3, k_3 e_4, r, 0.5) - z_{33}] \end{cases} \tag{23}$$

where $\text{fhan}(\cdot)$ is nonlinear function, as in [17]. k_3 and r are parameters. Note that the control law only consists of state error and designed parameters, this is because ESO has shown advantage that it requires few plant information. Once the control model is transformed to the normal ADRC model, the stability analysis of ADRC controller can be proved by Hurwitz criterion or Lyapunov method [24][25].

IV. SIMULATION

In this section, we first present the control authority and dynamic coupling of the moving mass. Then, the numerical simulation results of the proposed adaptive longitudinal controller applied to the moving mass flight vehicle model are presented. The parameters of the flight vehicle are chosen as m_S = 1000kg, $\mu_P = 0.5$, S = 3.8m^2, L = 4.11m, L_P = 2.08m, L_B = 2.2m, I_{P3} = 380kg·m^2, and I_{B3} = 440kg·m^2. According to the typical reentry mission profile, the flight height and velocity of reentry vehicles are 20km and 4080m/s, respectively [6]. Moreover, the simulation is assumed to execute at the characteristic point, which means the flight parameters are fixed.

A. Simulations of the dynamic analysis

Fig. 3 illustrates the coupling index between the deflection angle and the pitch channel for different mass ratios. Note that more mass ratio lead to curves change from concave to convex. An interesting aspect of this result is that when the mass ratio is close to 0.7, the coupling index curve becomes an approximate straight line. The coupling index is meaningful, since it can be used for designing the moving mass parameters to control attitude efficiently.

According to the Eq.(5), the curves of the trim AOA with different parameters can be illustrated in Fig. 4 and Fig. 5. Fig. 4 shows the effect of Δ_{BP} on the trim AOA. An interesting aspect of this result is that more control authority can be gained from a further Δ_{BP}, resulting from the decreased system static margin.

Fig. 5 shows the effect of the mass ratio on the trim AOA. Particularly, the trim AOA is linearly proportional to the mass ratio when $L_P = L_B$. The trim AOA increases slightly when $L_P < L_B$ and increases obviously when $L_P > L_B$ with increasing mass ratio. The reason is that increasing the mass ratio increases the static margin when $L_P < L_B$ and it is contrary when $L_P > L_B$.

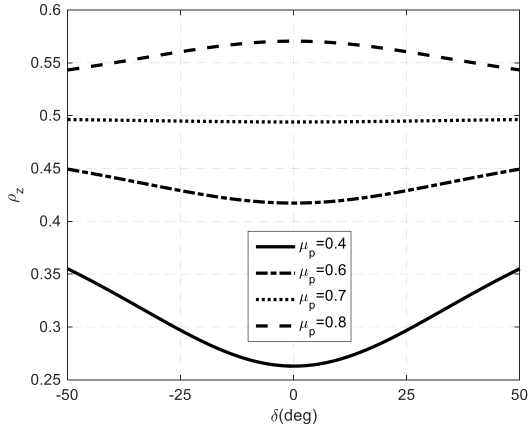

Fig. 3 the deflection angle and pitch channel coupling

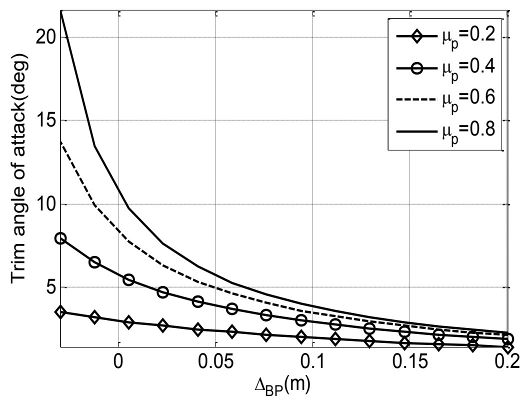

Fig. 4 Trim AOA vs Δ_{BP} for different mass ratio

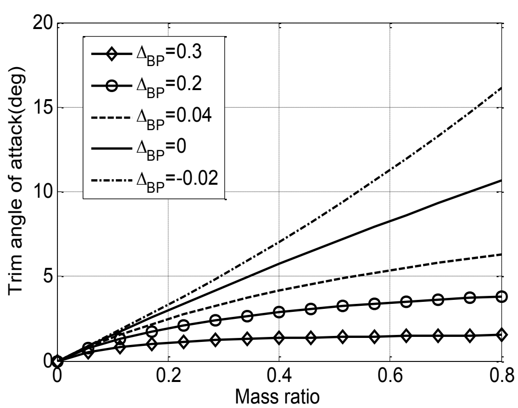

Fig. 5 Trim AOA vs mass ratio for different Δ_{BP}

B. *Simulations of the adaptive controller*

In this case, the parameters of the controller and observer are chosen by a tuning method based on bandwidth [21]. Thus, $k_1 = 10$, $\beta_{11} = 10$, $\beta_{12} = 25$, $k_2 = 30$, $\beta_{21} = 20$, $\beta_{22} = 100$, $k_3 = 80$, $\beta_{31} = 30$, $\beta_{31} = 30$, $\beta_{32} = 300$, $\beta_{33} = 1000$ in each ADRC. To verify the disturbance rejection ability of the controller, aerodynamic coefficients are perturbed with 30% uncertainty.

The command tracking trajectory of the angle of attack can be found in Fig. 6. It can be seen that the ADRC controller ensures a rapid convergence to the desired trajectories and tracks the commanded angles accurately, which shows good dynamic performance and steady-state performance when dealing with the uncertainty of aerodynamic parameters,

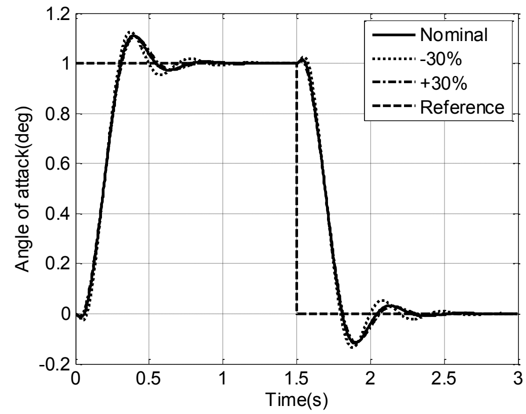

Fig. 6 Reference command tracking

Fig. 7 and Fig. 8 shows the deflection of the moving mass and the control input. Due to the great difference at the initial time, the control input vibrates obviously and becomes smooth later. In addition, the trim AOA is generated by the aerodynamic drag, the magnitude of the control input varies with aerodynamic uncertainty.

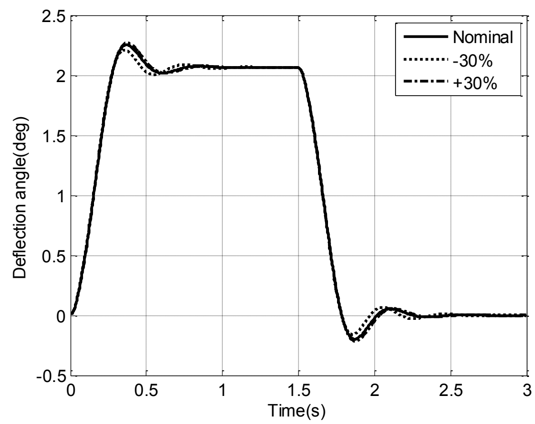

Fig. 7 Deflection angle of moving mass

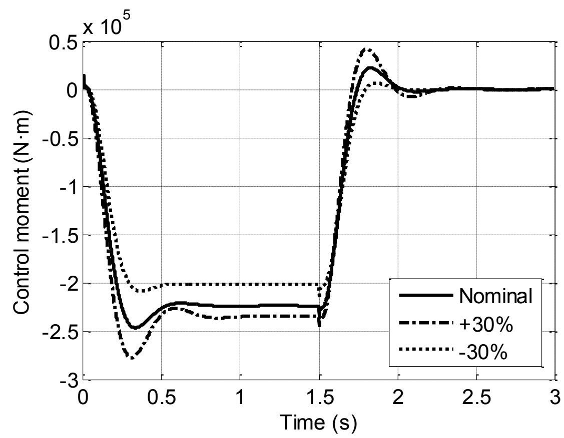

Fig. 8 control moment

The estimation errors of the total disturbance are shown in Fig. 9. It can be seen that the estimation errors converge to zero, which means the estimates converge to their corresponding true values.

The simulation results indicate that the proposed moving mass control mechanism can produce significant control authority for the flight vehicle. Moreover, the proposed adaptive controller solves the problem of the coupling influence of a moving mass with a large inertia on the attitude of the vehicle.

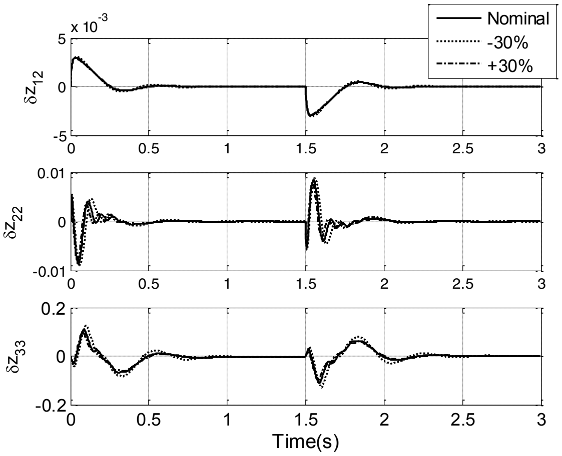

Fig. 9 estimation error of ESO

V. CONCLUSION

This paper proposes a novel moving mass configuration of flight vehicles and presents an adaptive longitudinal controller for moving mass vehicles based on the active disturbance reject control. The proposed configuration of moving mass with large mass ratio results in a close coupling between the body attitude and the actuator mass dynamics equations. This coupling is closely related to the moments of inertia of the moving mass. The steady-state trim angle of attack can be achieved by choosing a reasonable parameter value. The control authority can be increased with increasing moving-mass ratio or decreasing the mass center difference. To design the adaptive longitudinal controller, a cascade control strategy based on ADRC is presented. The coupling factors, unknown parameters and uncertainty are estimated using an extended state observer and are compensated by feedback control. Representative simulations are performed whose results show that the adaptive control law can achieve good command tracking with uncertain aerodynamic parameters.

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